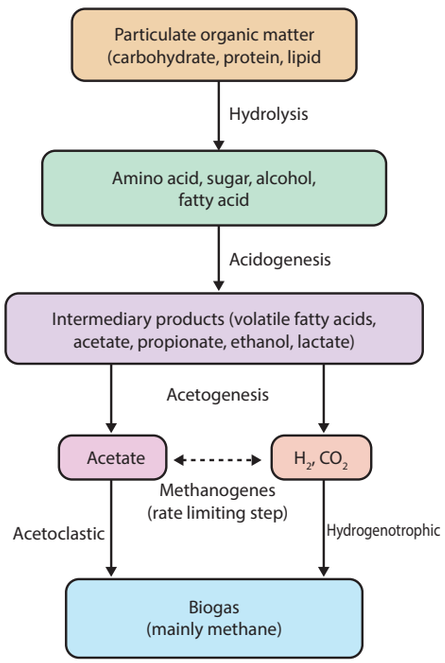
Acetogenic stage

Through various fermentation reactions, the products from hydrolysis are converted mainly into various organic acids (acetic acid, propionic acid, butyric acid, succinic acid, lactic acid), alcohols, ammonia (from amino acids), carbon dioxide and hydrogen. Facultative anaerobes and hydrogen producing bacteria Example: *Acetovibrio cellulosolvens, Bacteroid cellulosolvens* are involved.



Methanogenic stage

In this step, obligate anaerobic methane producing bacteria produce Methane gas as the major end product along with Carbon dioxide, Hydrogen and traces of other gases. Methanogenesis has six major pathways, each converting a different substrate into Methane gas. The six major substrates used are Carbon dioxide, Formic acid, Acetic acid, Methanol, Methylamine, and Dimethyl sulphate. The methanogenic bacteria include *Methanococcus voltae* and *Methanobacterium formicum* (Figure 9.14 a, b).



Deenabandhu model

It is a biogas production model popular in India which means "Friend of the helpless"

Small scale biogas unit

The biogas production is carried out in an air tight cylindrical tank called biogas digester (Figure 9.15).

Applications

1. Biogas used as fuel
2. Used to generate electricity
3. Biogas is used to run any type of heat engine in order to generate electrical and mechanical power.
4. Producing high quality fertilizer.
5. Reducing water and air pollution.

Summary

Environmental microbiology is the field of science that examines the relationship between microorganism and their biotic and abiotic environments.

Identification of new microbes and their products have practical application on protecting the environment as well as human health. Areomicrobiology is the study of airborne microorganisms and is one of the important modes for the transmission of infectious diseases. The air in our atmosphere is composed of different gaseous molecules. The air present both interior and exterior of the environment is called indoor air and outdoor air. The microorganisms are discharged out in different forms which are grouped on the basis of their relative size and moisture content. They are aerosols, droplet, droplet nuclei and infectious dust. Hospital – acquired infection are also known

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